

NORTH SYDNEY GIRLS HIGH SCHOOL

EXTENSION 2 MATHEMATICS

HSC ASSESSMENT TASK

TERM 1 2006

Instructions:

- Time allowed: 70 minutes + 2 minutes reading time.
- This task is worth 18% of your HSC assessment
- Attempt all four questions
- Show all necessary working
- Marks may be deducted for incomplete or poorly arranged work
- Start each question on a NEW PAGE
- Write on one side of the page only

Question1 (10 marks)

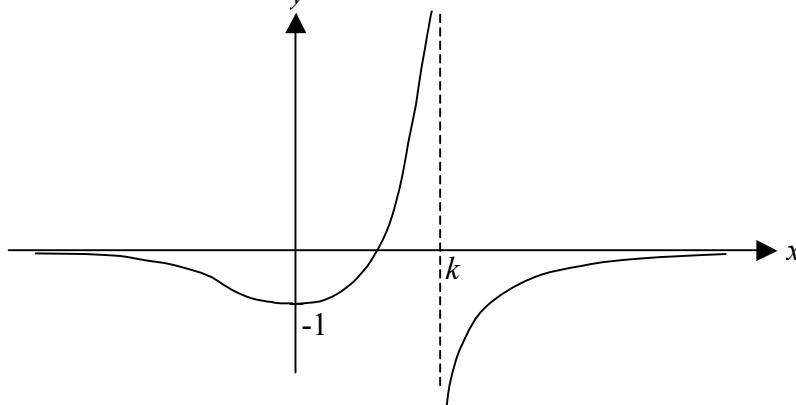
Marks

- (a) Given the complex number $z = 1 - \sqrt{3}i$ write down
- | | | |
|-------|----------|---|
| (i) | $ z $ | 1 |
| (ii) | $\arg z$ | 1 |
| (iii) | z^3 | 1 |
- (b)
- | | | |
|------|---|---|
| (i) | Solve the equation $z = \sqrt{5-12i}$ | 2 |
| (ii) | Given that $w = \frac{1 \pm \sqrt{5-12i}}{2+2i}$ is purely imaginary find w^{400} | 2 |
- (c)
- | | | |
|------|---|---|
| (i) | On the same Argand diagram, sketch the locus of
$ z-3 =1$ and $ z = z-2i $ | 2 |
| (ii) | Hence write down the complex number represented
by the intersection of the two loci. | 1 |

Question 2 (10 marks)

Marks

The graph of $y = f(x)$ is shown below



Draw neat sketches of the following

- | | | |
|-------|----------------|---|
| (i) | $y = f(x - k)$ | 2 |
| (ii) | $y = f(-x)$ | 2 |
| (iii) | $y = f(x)$ | 2 |
| (iv) | $y = [f(x)]^2$ | 2 |
| (v) | $y = e^{f(x)}$ | 2 |

Question 3. (10 marks)

Marks

- (a) A conic is defined by the equation $\frac{x^2}{19-k} + \frac{y^2}{7-k} = 1$
- (i) Determine the values of k for which the equation defines an ellipse 1
- (ii) Consider the above conic when $k = 3$. Write down the value of the eccentricity, the coordinates of the foci and the equations of the directrices. 3
- (iii) Sketch the above ellipse ($k = 3$) showing all important features. 2
- (iv) Describe how the shape of the ellipse changes as k increases in value from 3 to 7. 1
- (b) Write down the equations of the asymptotes to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. 3

Hence deduce that the eccentricity $e = \sqrt{2}$ if these asymptotes are perpendicular to each other.

Question 4. (10 marks)

- (a) The rectangle $OABC$ lies in the Argand plane where OC is twice OA . Vertex O corresponds to the origin.
The vertex A corresponds to the complex number w .
Express in terms of w , the complex number that corresponds to
- (i) the point C 1
- (ii) the point D , the point of intersection of the diagonals. 2
- (b) z is a variable complex number represented by the point P . Describe and sketch the locus of P if $|z - 4i| = \text{Im}(z)$ 4
- (c) Let $z = \cos(\frac{\pi}{4}) + i \sin(\frac{\pi}{4})$ and $w = \cos(\frac{\pi}{3}) + i \sin(\frac{\pi}{3})$. Evaluate zw and hence give the exact value of $\cos(\frac{7\pi}{12})$ 3

Question 5. (10 marks)

Marks

- (a) Draw a neat sketch of $y = |x - 4| + |x|$ 2
- (b) On the same set of axes draw a neat sketch of $y = \cos x$ and $y = \ln x$ for $0 \leq x \leq 2\pi$. 4
- Hence draw a neat sketch of $y = \cos x \ln x$ for $0 \leq x \leq 2\pi$.
- (c) Draw a neat sketch of $y = x^2(1 - x^2)$ showing all intercepts and the coordinates of any stationary points. 4
- Hence sketch the relation $y^2 = x^2(1 - x^2)$

Question 6. (10 marks)

- (a) Explain why the set of points z in the Argand plane defined by $|z + 5| - |z - 5| = 8$ defines part of a conic. 4
- Sketch this locus showing essential features.
- (b) Consider the two series, C and S where
 $C = 1 + \cos \theta + \cos 2\theta + \dots \dots \dots \cos(n-1)\theta$ and
 $S = \sin \theta + \sin 2\theta + \dots \dots \dots + \sin(n-1)\theta$
- (i) Write down $C + iS$ in terms of θ 1
- (ii) If $z = \cos \theta + i \sin \theta$, express $C + iS$ as a series in terms of z . 3
- Hence show that $C + iS = \frac{1 - z^n}{1 - z}$, $z \neq 1$
- (iii) Show that 2
- $$1 + \cos \theta + \cos 2\theta + \dots \dots \dots \cos(n-1)\theta = \frac{\cos[(n-1)\theta] - (\cos \theta + \cos n\theta) + 1}{2 - 2 \cos \theta}$$

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**End of paper**

# Solutions

Question 1 (19 marks)

(a) (i)  $|z| = \sqrt{1+3} = 2$

(ii)  $\arg z = -\frac{\pi}{3}$

(iii)  $z^3 = \left[ 2 \left( \cos\left(\frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{3}\right) \right) \right]^3$   
 $= 8 \left[ \cos(-\pi) + i\sin(-\pi) \right]$   
 $= -8$  (2)

(b) (i) let  $z = x+iy$   
 $\therefore x^2 - y^2 = 5, \quad 2xy = -12$

$\therefore x = 3, y = -2$

$\therefore \sqrt{5-12i} = 3-2i$  (3)

(ii)  $w = \frac{3 \pm \sqrt{5-12i}}{2}$

$= \frac{3 \pm (3-2i)}{2}$

$= \frac{6-2i}{2}, \frac{2i}{2}$

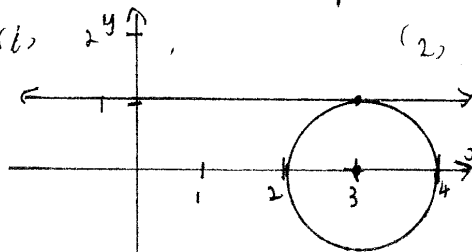
$= 3-i, i$

But  $w$  imaginary

$\therefore w = i$

$\therefore w^{400} = i^{400} = 1$  (2)

(c) (i)  $z = 3+i$  (1)



(ii)  $z = 3+i$

(d)  $|z - 4i| = \operatorname{Im}(z)$

let  $z = x+iy$

$|x + (y-4)i| = y$

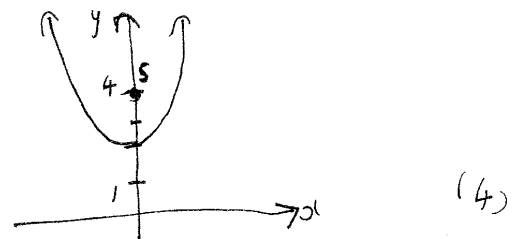
$\sqrt{x^2 + (y-4)^2} = y$

$x^2 + y^2 - 8y + 16 = y^2$

$x^2 = 8y - 16$

$x^2 = 8(y-2)$

Parabola, concave up vertex (0, 2) axis of



(e)  $z = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}(1+i)$

$w = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3} = \frac{1}{2}(1+\sqrt{3}i)$

$zw = \left( \cos\left(\frac{\pi}{4} + \frac{\pi}{3}\right) + i\sin\left(\frac{\pi}{4} + \frac{\pi}{3}\right) \right)$

$= \frac{1}{2\sqrt{2}}(1+i)(1+\sqrt{3}i)$

$= \frac{1}{2\sqrt{2}}(1+i+\sqrt{3}i-\sqrt{3})$

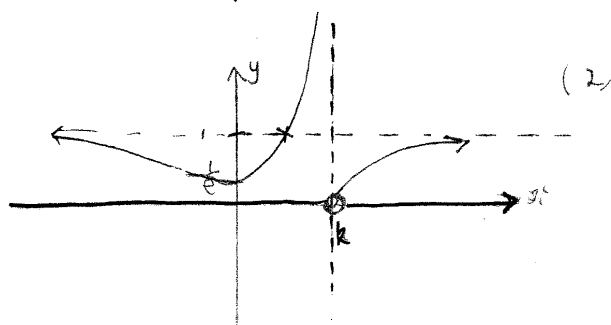
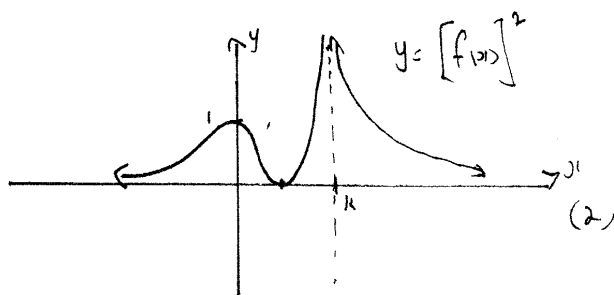
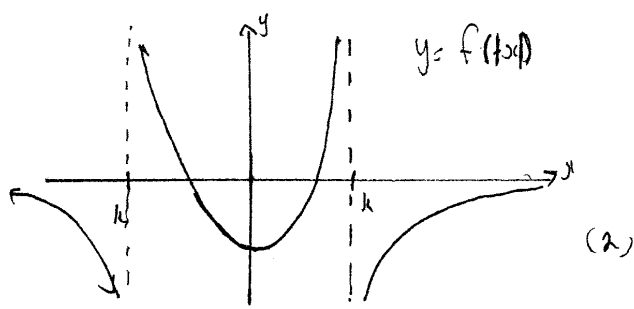
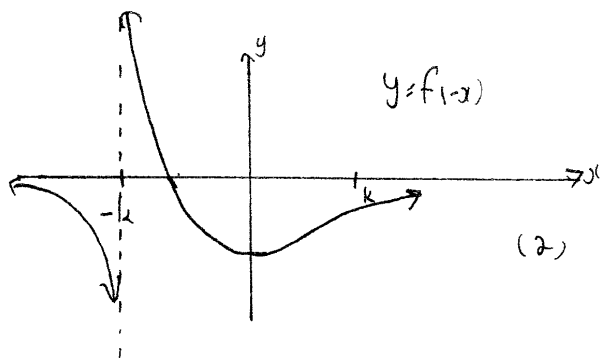
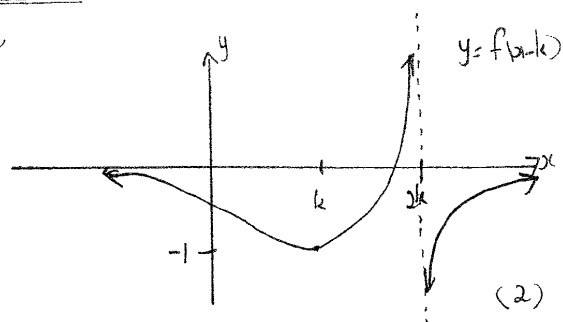
$\therefore \cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12} = \frac{1-\sqrt{3}}{2\sqrt{2}} + \frac{(1+\sqrt{3})i}{2\sqrt{2}}$

Comparing real parts

$\cos\frac{7\pi}{12} = \frac{1-\sqrt{3}}{2\sqrt{2}}$  (3)

## Question 2 (18 marks)

(a)



(b)  $y = x^2(1-x^2)$  Even

Intercepts  $x = 0, \pm 1$

$$y = x^2 - x^4$$

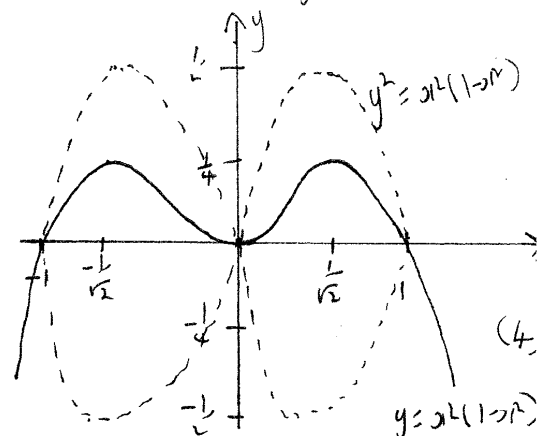
$$y' = 2x - 4x^3 = 2x(1-2x^2)$$

$$y'' = 2 - 12x^2$$

Spl's  $x = 0, y = 0$   $y'' > 0 \therefore \text{Min}$

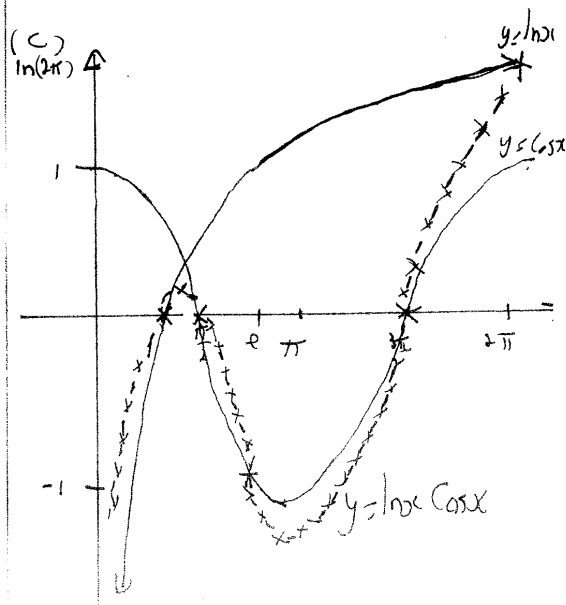
$x = \frac{1}{\sqrt{2}}, y = \frac{1}{4}$   $y'' < 0$  Max

$x = -\frac{1}{\sqrt{2}}, y = \frac{1}{4}$   $y'' < 0$  Max.



$$y^2 = x^2(1-x^2)$$

$$y = \pm \sqrt{x^2(1-x^2)}$$

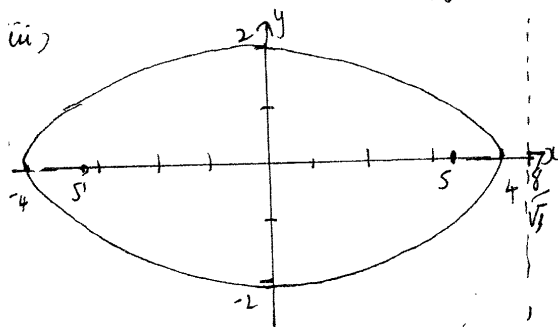


### Question 3 (20 marks)

(a),  $19-k > 0$  and  $7-k > 0$   
 $k < 19$  and  $k < 7$  i.e.  $k < 7$

(ii),  $k=3 \rightarrow \frac{x^2}{16} + \frac{y^2}{4} = 1$   
 $a=4, b=2 \therefore 1-e^2 = \frac{b^2}{a^2} = \frac{1}{4}$   
 $e = \frac{\sqrt{3}}{2}$

$\therefore$  foci  $S(\pm 2\sqrt{3}, 0)$   
 vertices  $x = \pm a = \pm 4, y = \pm \frac{8}{\sqrt{3}}$

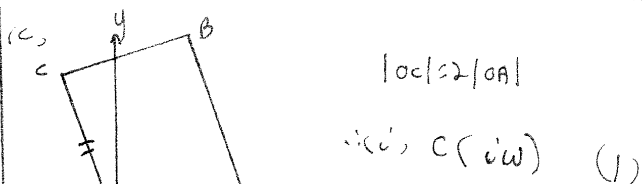
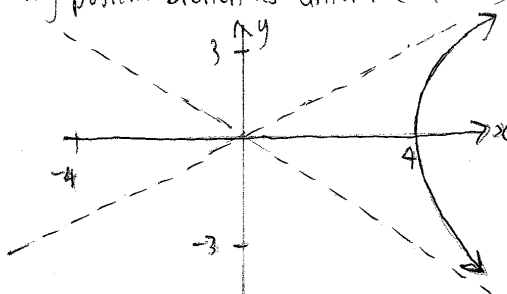


(i), as  $k \rightarrow 7$   $a \rightarrow \sqrt{7}, b \rightarrow 0$   
 gets flatter  $\rightarrow$  line interval  $SS'$ .

(b) Hyperbola since the difference in the distances to two fixed points is a constant.

Focus  $(\pm 5, 0)$   $2a=8 \therefore a=4$   
 $\therefore ae=5 \therefore e=\frac{5}{4} \therefore b=3$

$\frac{x^2}{16} - \frac{y^2}{9} = 1$  asymptotes  $y = \pm \frac{3}{4}x$   
 only positive branch as difference  $> 0$



(ii), D midpoint of AC i.e.  $D(\frac{zw+w}{2})$   
 or  $w(\frac{z+1}{2})$

(d)  $z^n = 1 + (z + z^2 + \dots + z^n) + z^{n+1} + \dots$   
 $\dots = z^n [1 + z + z^2 + \dots + z^{n-1}]$

(ii),  $\therefore z^n = 1 + (z + z^2 + \dots + z^n) + z^{n+1} + \dots$

$= 1 + z + z^2 + \dots + z^{n-1}$   
 -g.p.  $a=1, r=z, n=n$

$\therefore z^n = \frac{1(1-z^n)}{1-z} = \frac{1-z^n}{1-z}, z \neq 1$

(iii),  $z^n = \frac{1 - (z^n + z^{n+1} + \dots)}{1 - (z + z^2 + \dots)}$   
 $= \frac{[1 - z^n] - z^{n+1}}{[1 - z] - z^{n+1}} \times \frac{[1 - z^n] + z^{n+1}}{[1 - z] + z^{n+1}}$   
 $= \frac{(1 - z^n)(1 - z^n) + z^{n+1}z^n - z^{n+1}(1 - z^n) - z^{n+1}z^n}{1 - 2z^n + z^{2n} + z^{2n+1}}$

Comparing real parts

$C = \frac{(1 - \cos \theta)(1 - \cos n\theta) + \sin n\theta \sin \theta}{2 - 2 \cos \theta}$   
 $= \frac{1 - \cos \theta - \cos n\theta + \cos n\theta \cos \theta + \sin n\theta \sin \theta}{2 - 2 \cos \theta}$

$1 + \cos \theta + \cos 2\theta + \dots + \cos(n-1)\theta$

$= \frac{\cos(n\theta - \theta) - (\cos \theta + \cos n\theta)}{2 - 2 \cos \theta}$   
 $= \frac{\cos(n-1)\theta - (\cos \theta + \cos n\theta)}{2 - 2 \cos \theta}$